SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Visualization Design Activity

Course Code:	DSA0502	**Course Title:**	Query Processing for Data Science
CO Assessed:	CO5	**Assessment:**	Tool 3– **Visualization Design Activity**
Weightage:	35%	**Total Marks:**	

CO5 Statement: Crea te meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication

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Section / Batch:	2023/AI&DS	**Date of Submission:**	20.08.26
Faculty In-charge:	**Dr. B.Shyamala Gowri**	**Project Mode:**	Individual Submission

Code of Conduct

I Sania Herbert(192324062) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: Sania

Distribution Detective
Que 1:Create a histogram of student marks or employee salaries. Identify the approximate shape, center, spread and possible skewness

Introduction

A **distribution** describes how values are spread across a dataset. A histogram is one of the most useful graphs for studying the distribution of numerical data.

Unlike a bar chart, a histogram groups continuous numerical values into intervals called **bins**. The height of each bar represents the number of observations falling within that interval.

Objective

To analyze the distribution of student marks using a histogram and identify its **shape, center, spread, and skewness**.

Interpretation

Shape

The histogram shows that the observations are concentrated mainly in the middle-to-high mark ranges.

Center

The mean is approximately **74.96**, while the median is **75**.

Therefore, the center of the distribution is approximately **75 marks**.

Spread

The lowest mark is **45** and the highest mark is **92**.

Therefore:

Spread = Maximum − Minimum

Spread = 92 − 45 = 47

Thus, the marks cover a range of **47 marks**.

Skewness

The mean and median are almost equal.

This suggests that the distribution is approximately symmetric and does not have strong skewness.

DATASET:

45, 52, 56, 60, 62, 65, 67, 68, 70, 72,
73, 74, 75, 76, 78, 80, 81, 82, 84, 85,
86, 87, 88, 90, 92

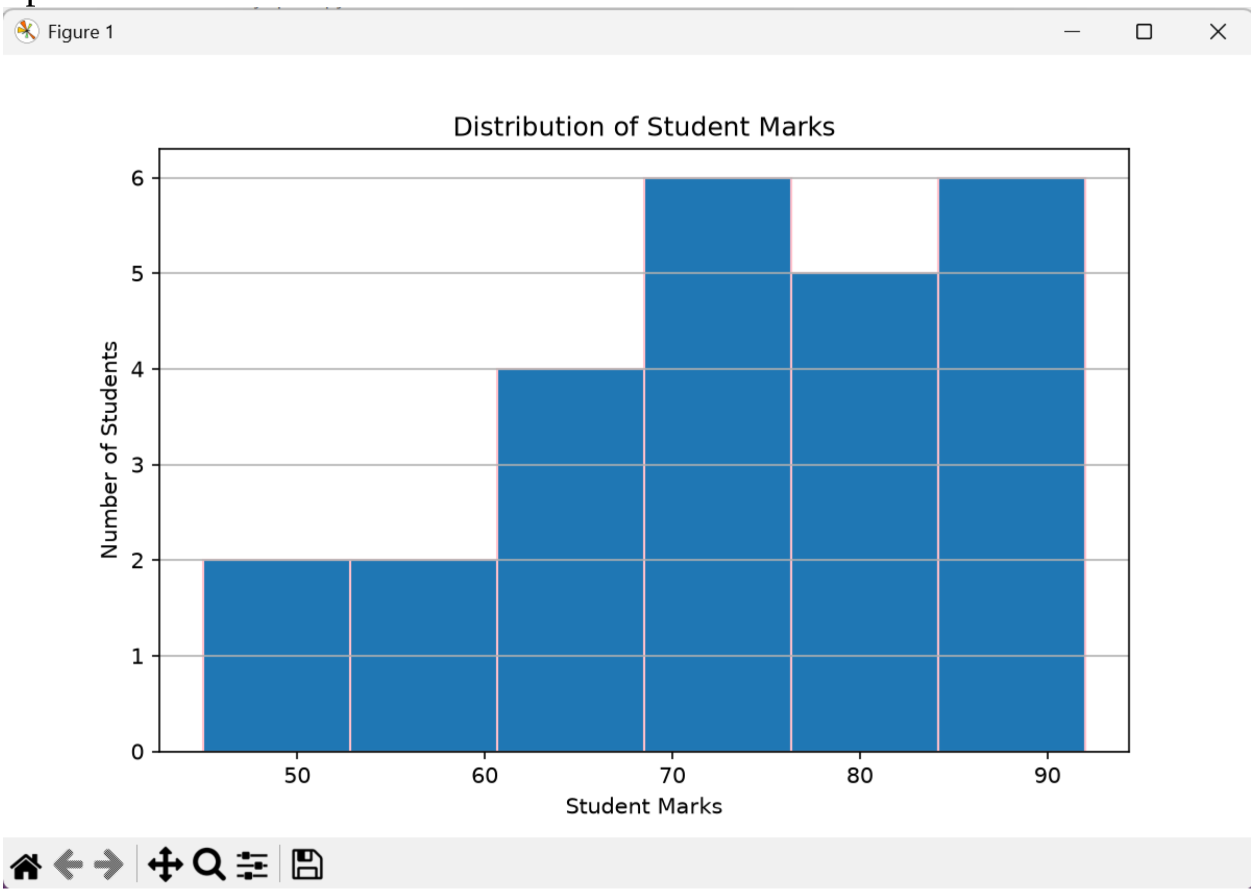
Python code:

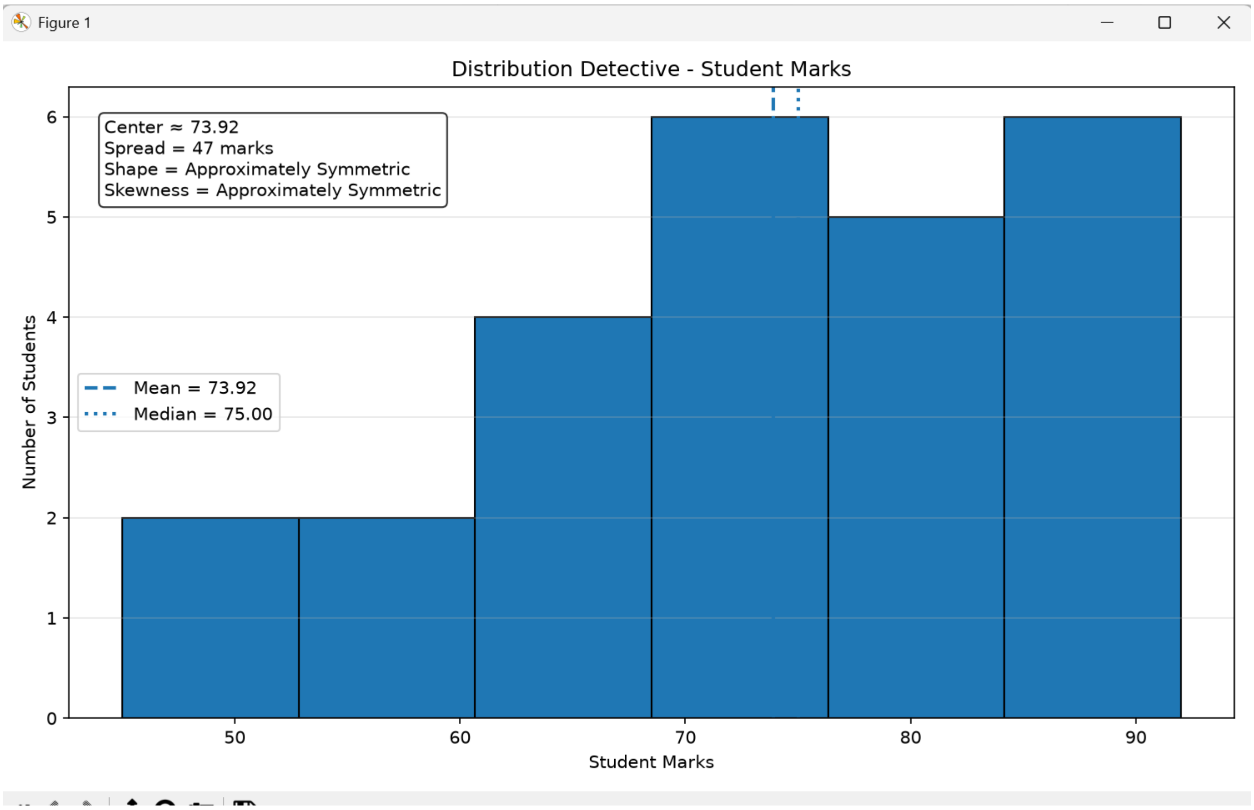
```
import pandas as pd
import matplotlib.pyplot as plt
# Student marks
marks = [
    45, 52, 56, 60, 62,
    65, 67, 68, 70, 72,
    73, 74, 75, 76, 78,
    80, 81, 82, 84, 85,
    86, 87, 88, 90, 92
]
df = pd.DataFrame({"Marks": marks})
# Calculate basic statistics
mean = df["Marks"].mean()
median = df["Marks"].median()
minimum = df["Marks"].min()
maximum = df["Marks"].max()
spread = maximum - minimum
print("Mean:", mean)
print("Median:", median)
print("Minimum:", minimum)
print("Maximum:", maximum)
print("Spread:", spread)
# Create histogram
```

```
plt.figure(figsize=(8, 5))
plt.hist(
    df["Marks"],
    bins=6,
    edgecolor="black"
)
plt.xlabel("Student Marks")
plt.ylabel("Number of Students")
plt.title("Distribution of Student Marks")
plt.grid(axis="y")
plt.show()
```

OUTPUT:

Mean: 73.92
Median: 75.0
Minimum: 45
Maximum: 92
Spread: 47





Analysis

The histogram shows that most student marks are concentrated around the middle-to-high range. The mean and median are almost equal, indicating that the distribution is approximately symmetric. There is no significant skewness.

Conclusion

The histogram provides a clear visual understanding of the student marks. Most students scored around **75 marks**, with a total spread of **47 marks**. The distribution is approximately symmetric.

Que:2 Lag Plot Investigation

Use time-series data to create a lag plot. Examine whether the current observation appears related to the previous observation.

Introduction

Time-series data consists of observations recorded in a particular sequence over time.

Examples include:

- Daily sales
- Monthly revenue
- Temperature
- Stock prices
- Website traffic
- Electricity consumption

Objective

To investigate whether the current observation in a time series is related to its previous observation using a **lag plot**.

Interpretation of the Lag Plot

The points show a general upward pattern.

This indicates that higher previous-day sales tend to be associated with higher current-day sales.

Therefore, the series demonstrates **positive temporal dependence**.

This is commonly described as **positive autocorrelation**.

Types of Lag Plot Patterns

Pattern	Interpretation
Upward pattern	Positive autocorrelation
Downward pattern	Negative autocorrelation
Random scatter	Little/no autocorrelation
Strong curved pattern	Possible nonlinear relationship

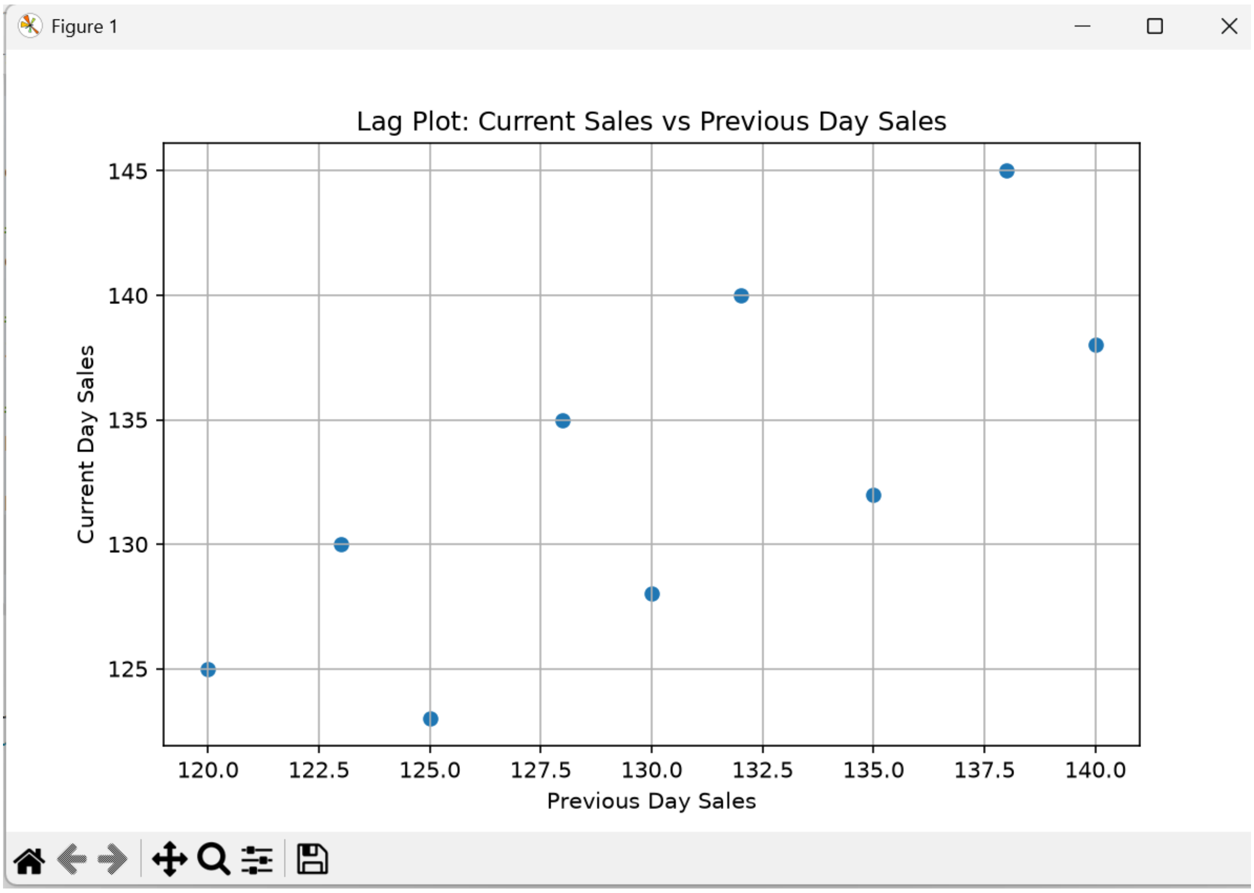
DATASET

Day	Sales
1	120
2	125
3	123
4	130
5	128
6	135
7	132
8	140
9	138
10	145

Python code:

```
import pandas as pd
import matplotlib.pyplot as plt
# Create time-series data
data = {
    "Day": range(1, 11),
    "Sales": [120, 125, 123, 130, 128, 135, 132, 140, 138, 145]
}
df = pd.DataFrame(data)
# Create lagged values
df["Previous_Day_Sales"] = df["Sales"].shift(1)
# Remove the first row because it has no previous value
lag_data = df.dropna()
# Create lag plot
plt.figure(figsize=(8, 5))
```

```
plt.scatter(
    lag_data["Previous_Day_Sales"],
    lag_data["Sales"]
)
plt.xlabel("Previous Day Sales")
plt.ylabel("Current Day Sales")
plt.title("Lag Plot: Current Sales vs Previous Day Sales")
plt.grid(True)
plt.show()
```



Types of Lag Pattern

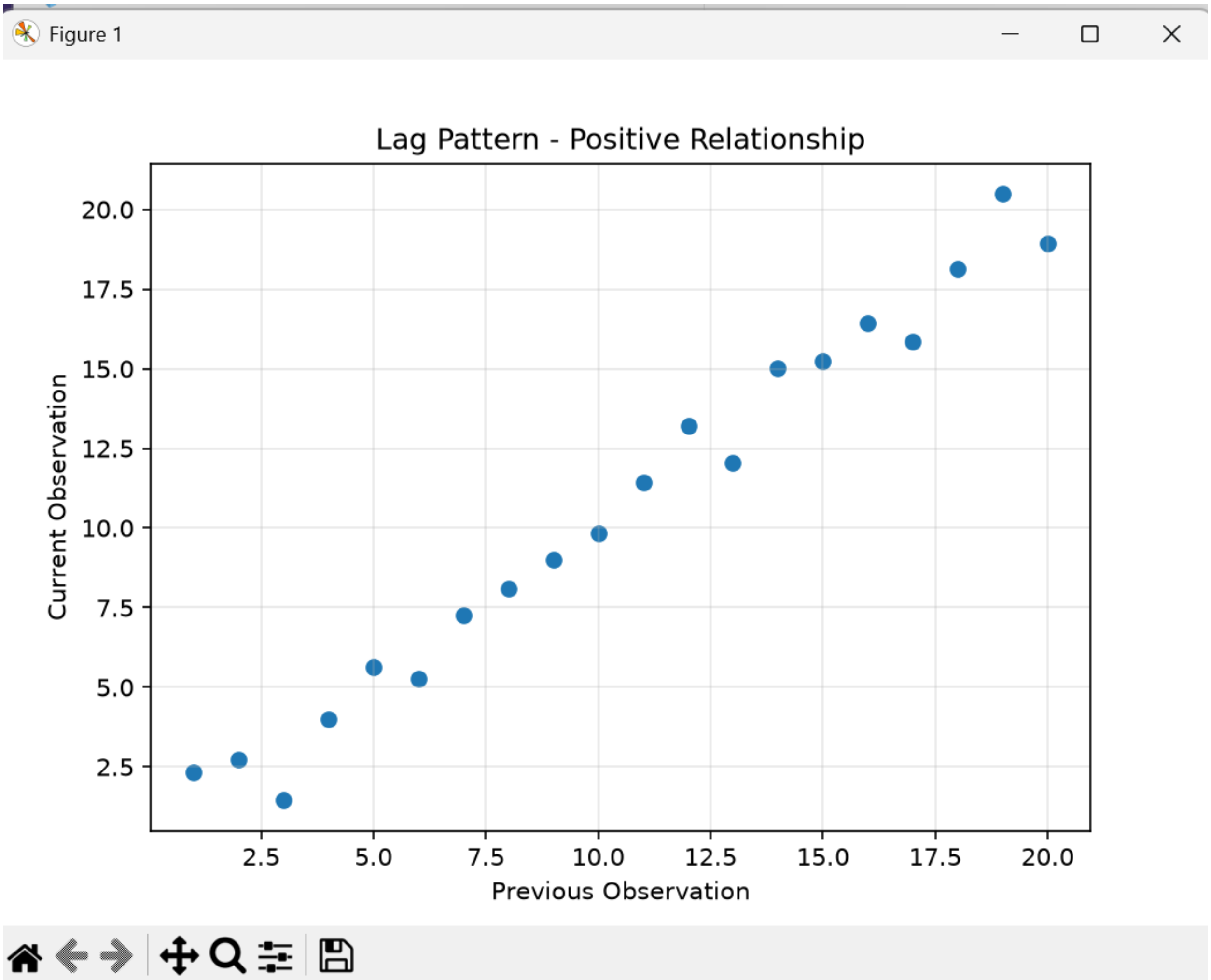


Figure 1
Lag Pattern - Positive Relationship
Current Observation
Previous Observation
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

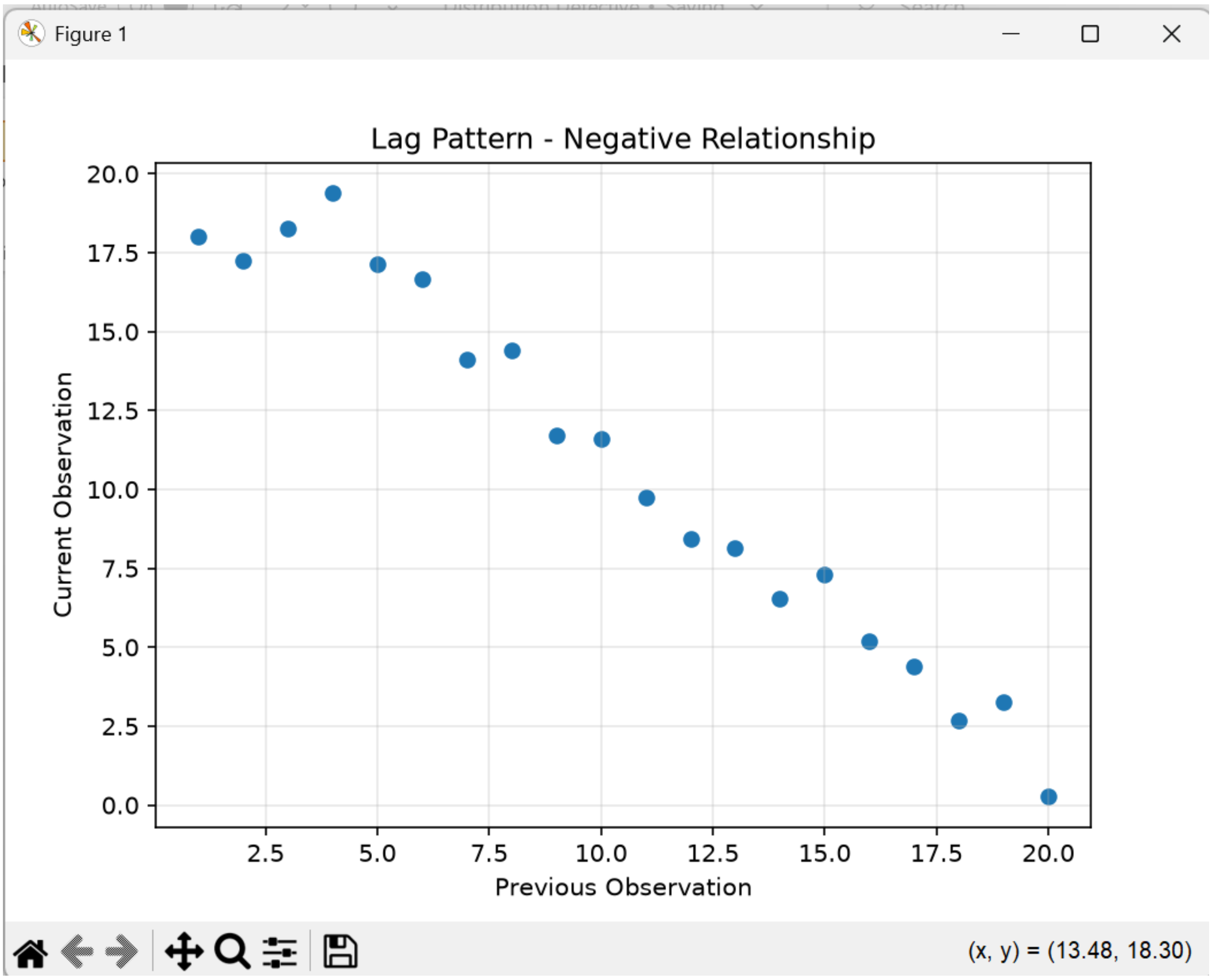
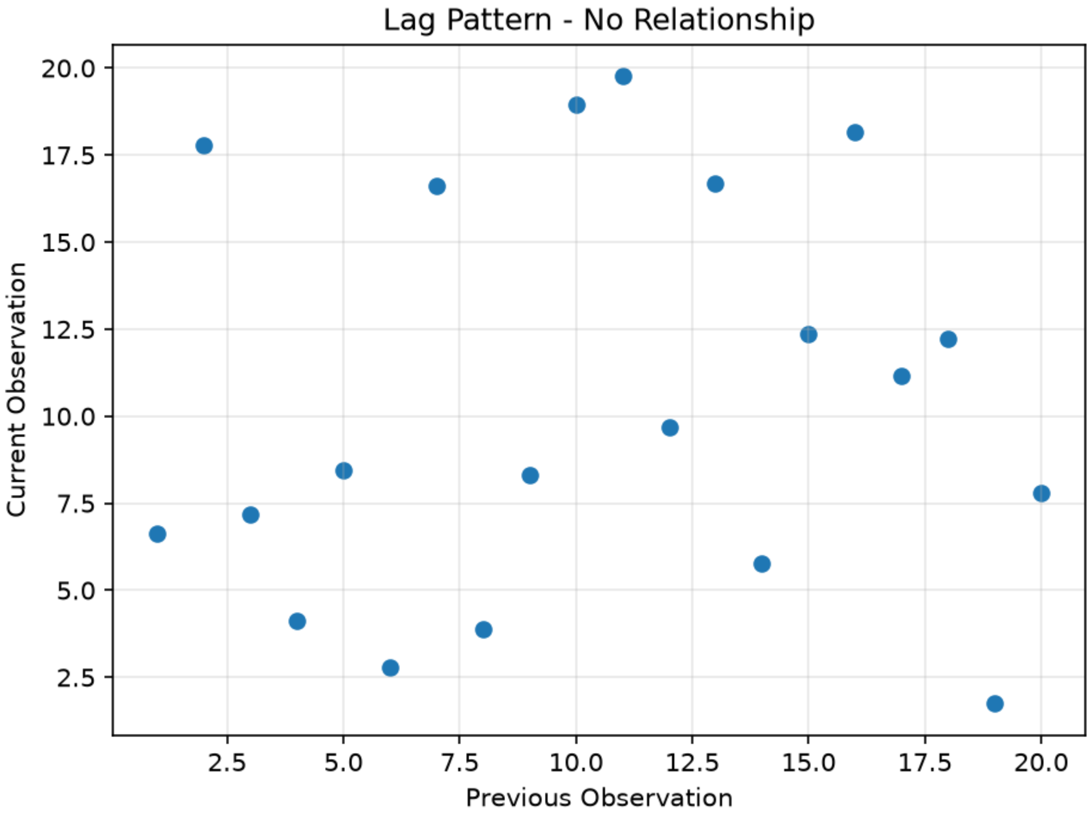


Figure 1
Lag Pattern - Negative Relationship
Current Observation
Previous Observation
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
(x, y) = (13.48, 18.30)



Lag Pattern - No Relationship
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Current Observation
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Previous Observation

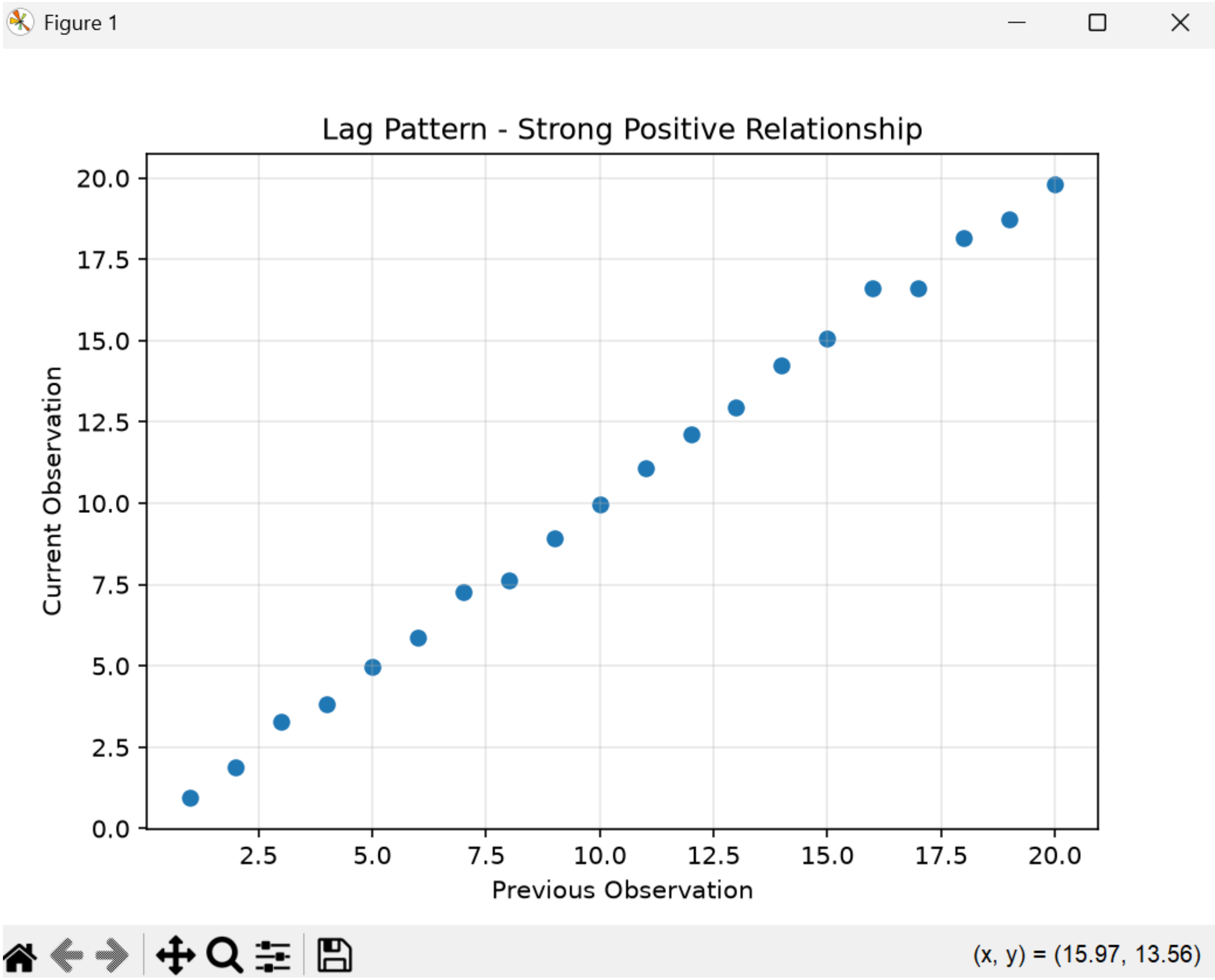


Figure 1
Lag Pattern - Strong Positive Relationship
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Current Observation
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Previous Observation
(x, y) = (15.97, 13.56)

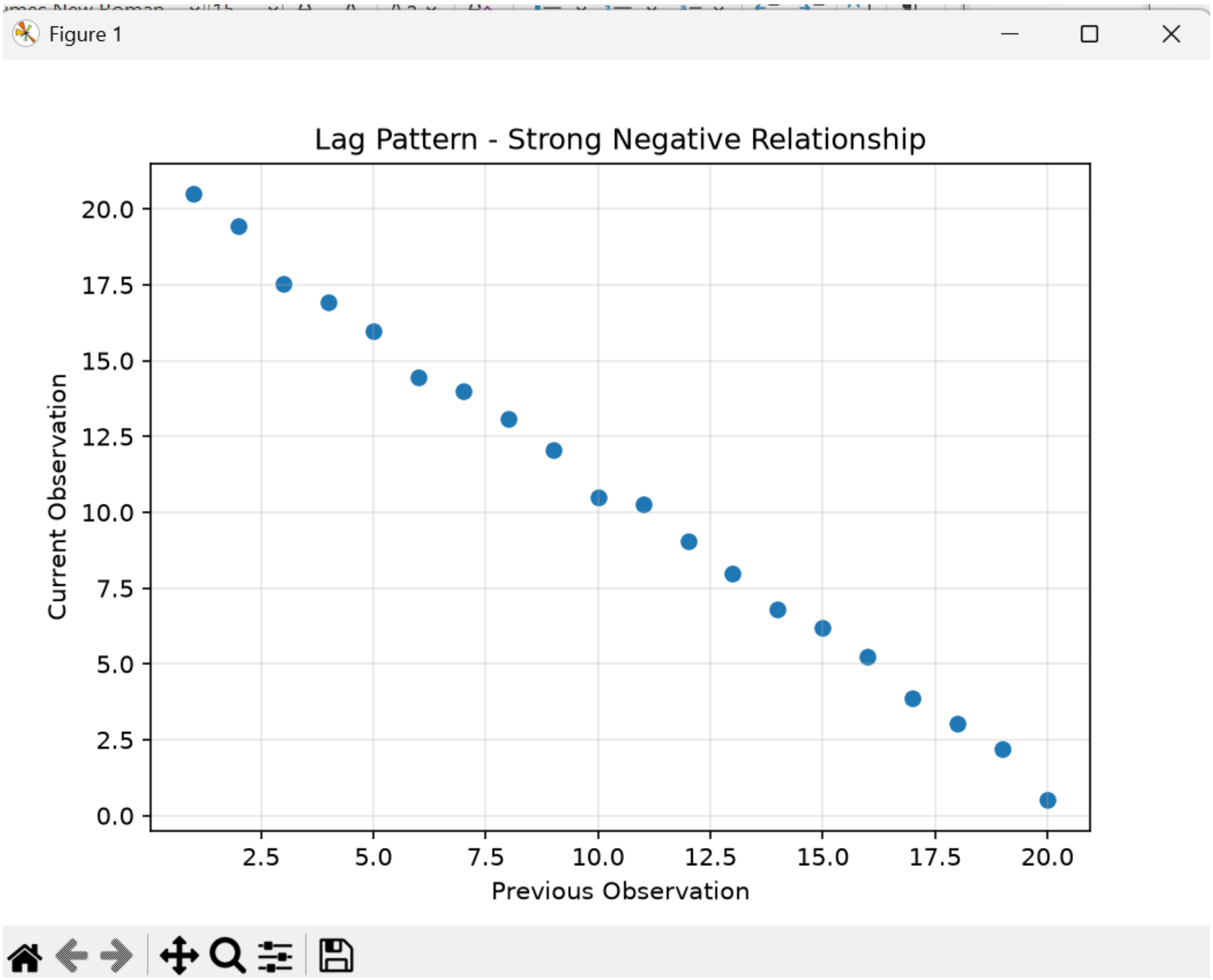
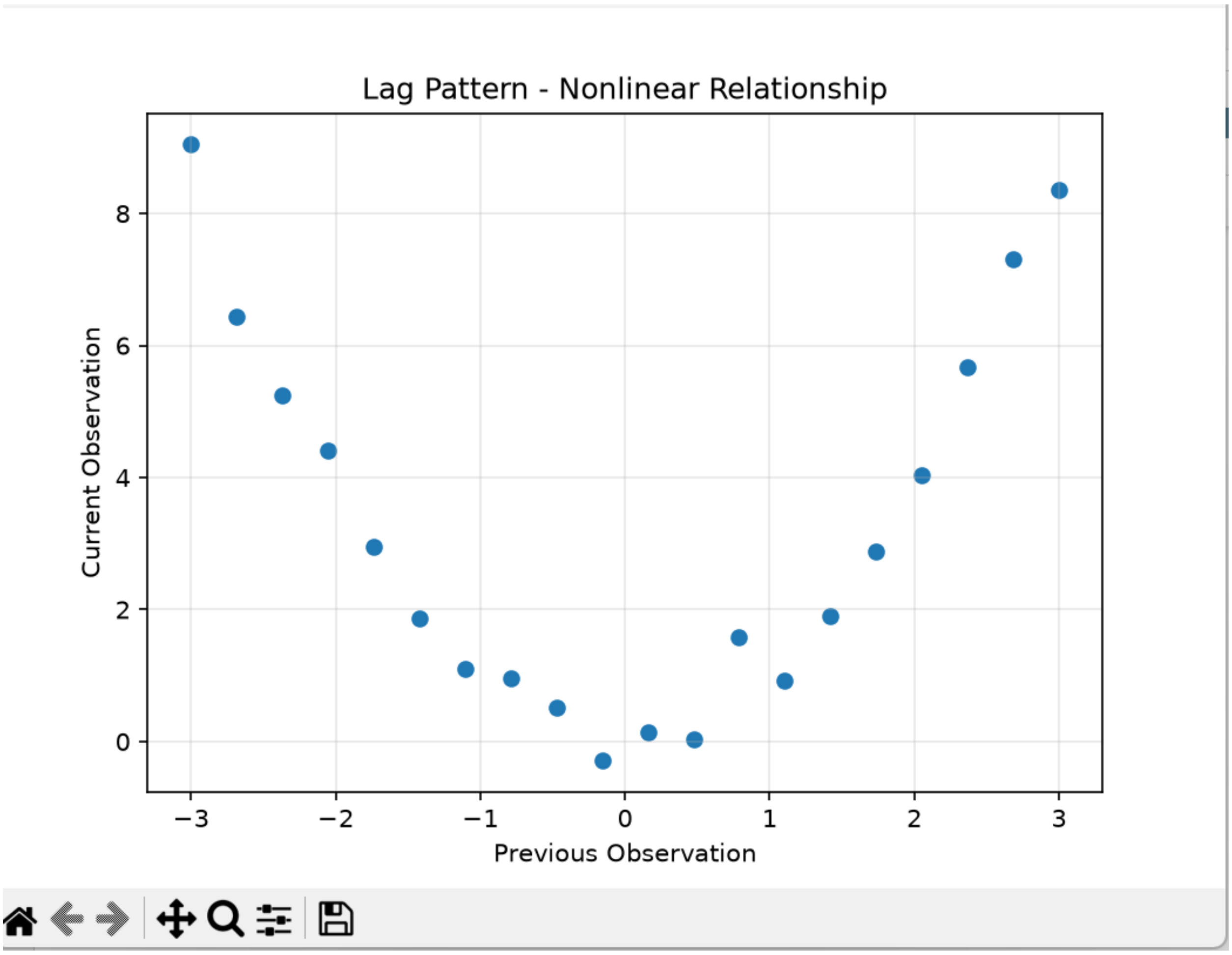


Figure 1
Lag Pattern - Strong Negative Relationship
Current Observation
Previous Observation
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0



Analysis

The points show an overall upward pattern. This means higher sales on the previous day tend to be associated with higher sales on the current day.

Therefore, the observations show **positive autocorrelation**.

Conclusion

The lag plot indicates that the current observation is related to the previous observation. This suggests that the sales time series contains temporal dependence rather than completely independent observations.